Audit Common Threats & Honeypots Report/Guide

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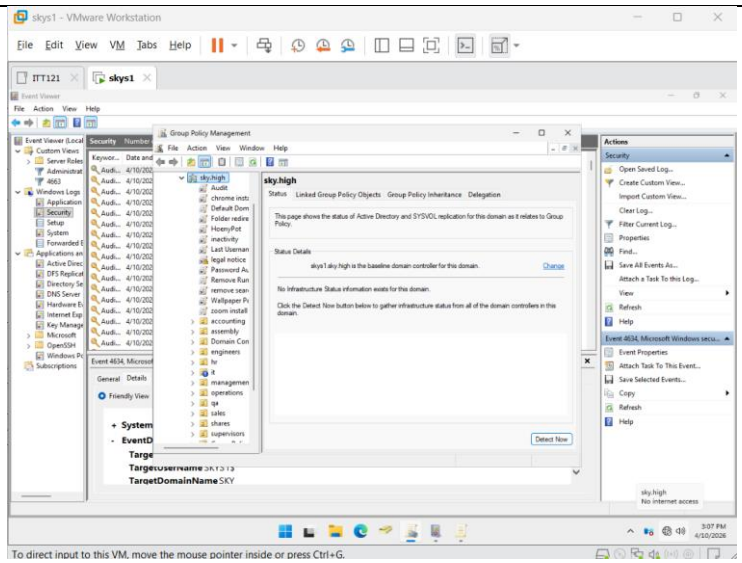
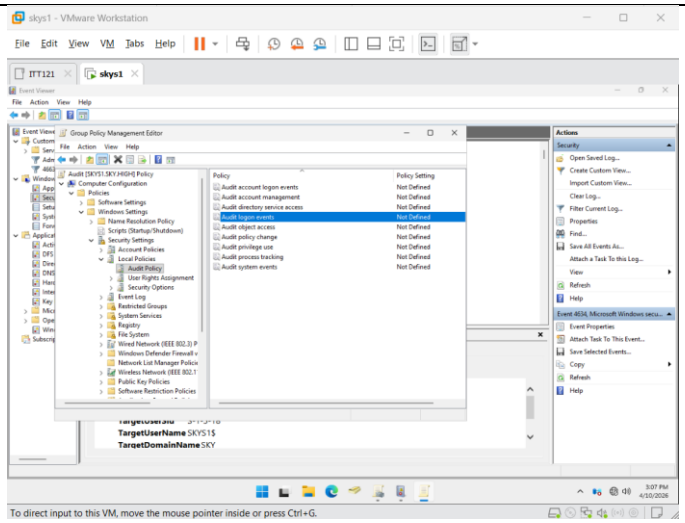
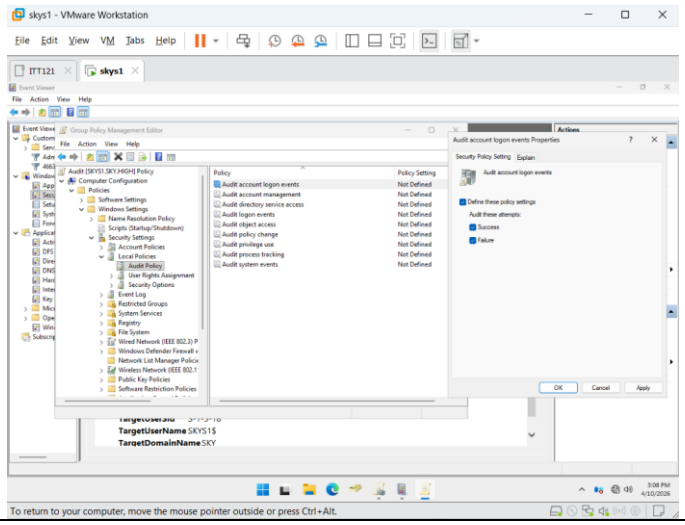
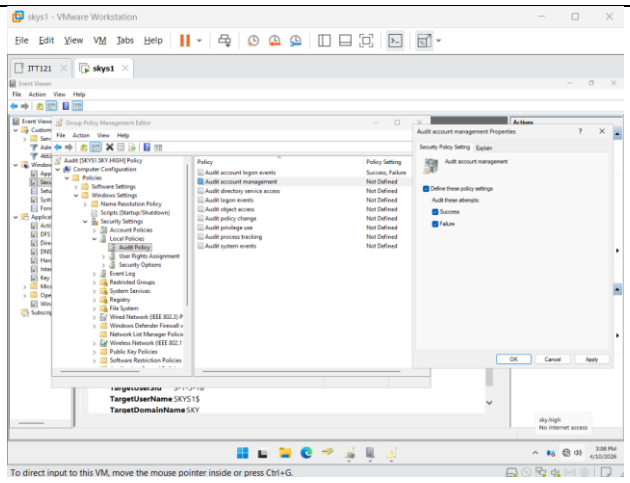
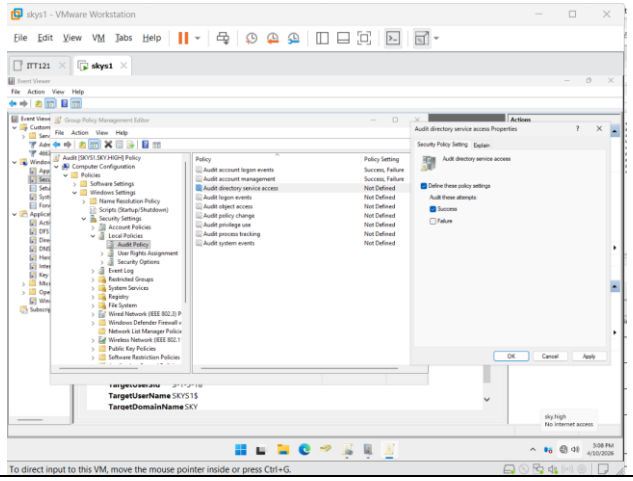
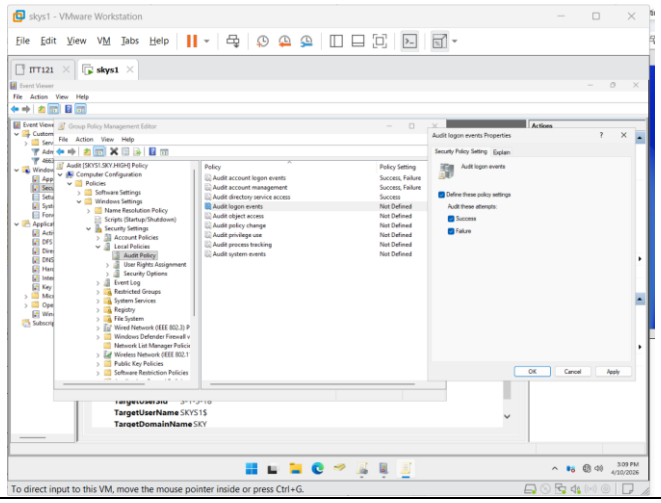
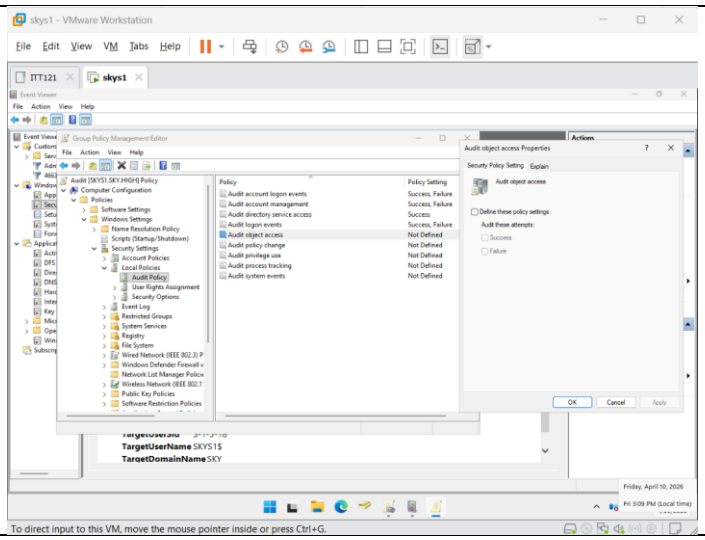
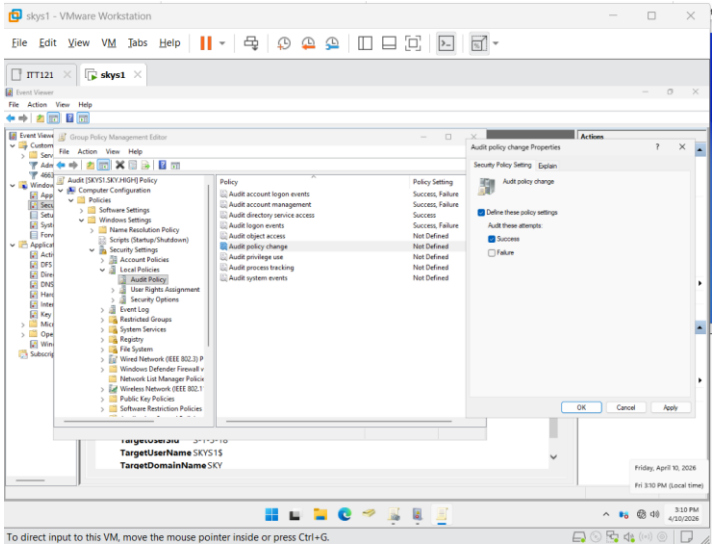
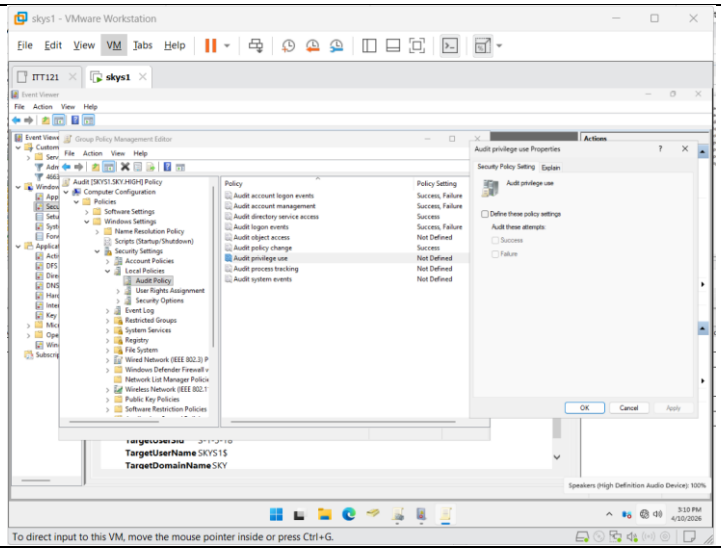
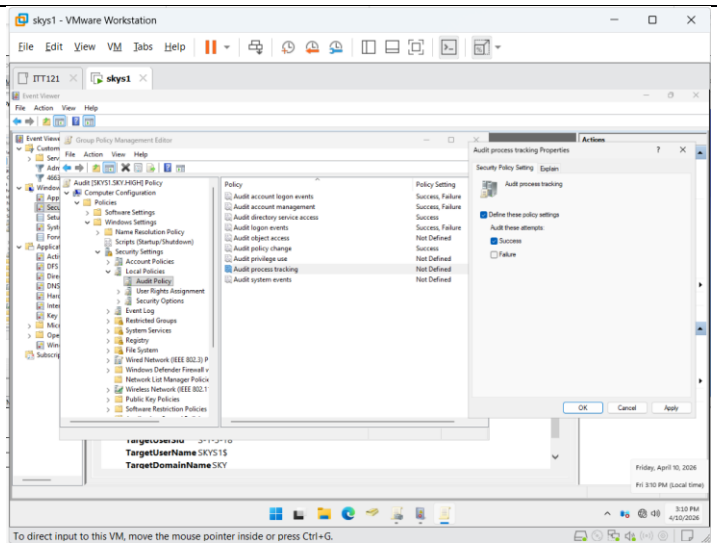
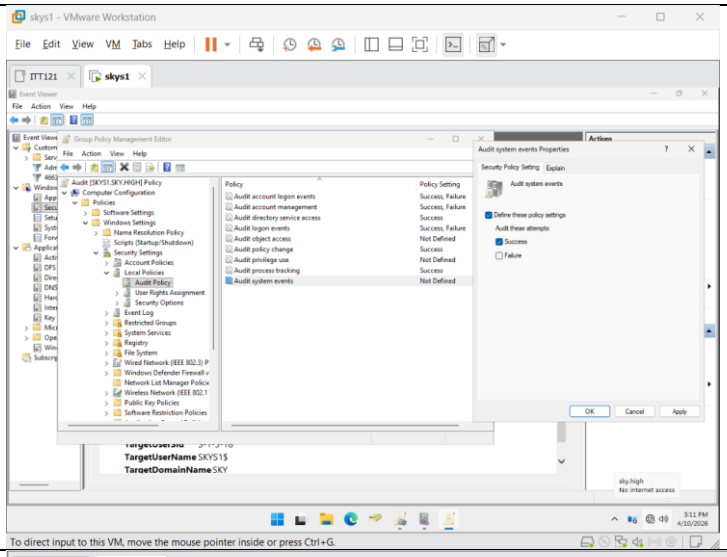
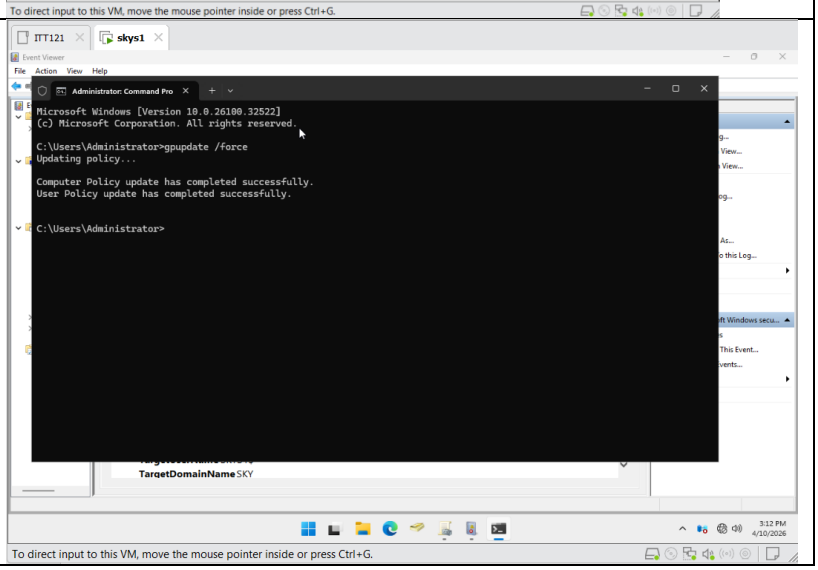
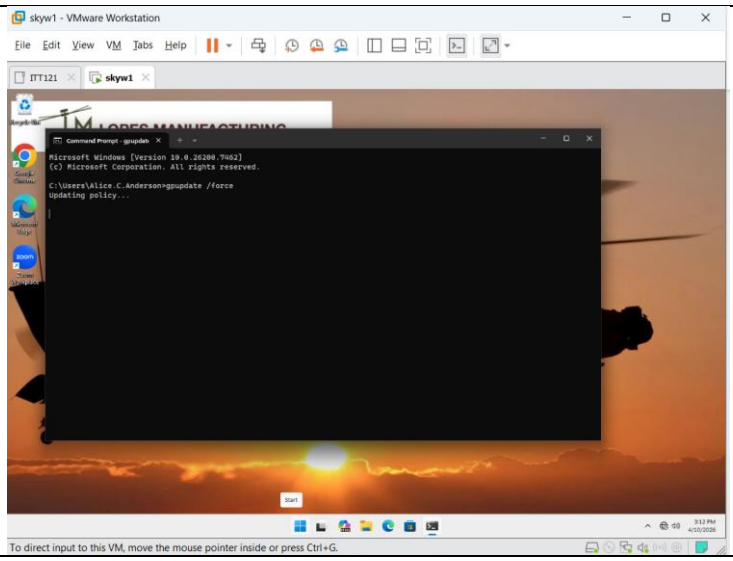
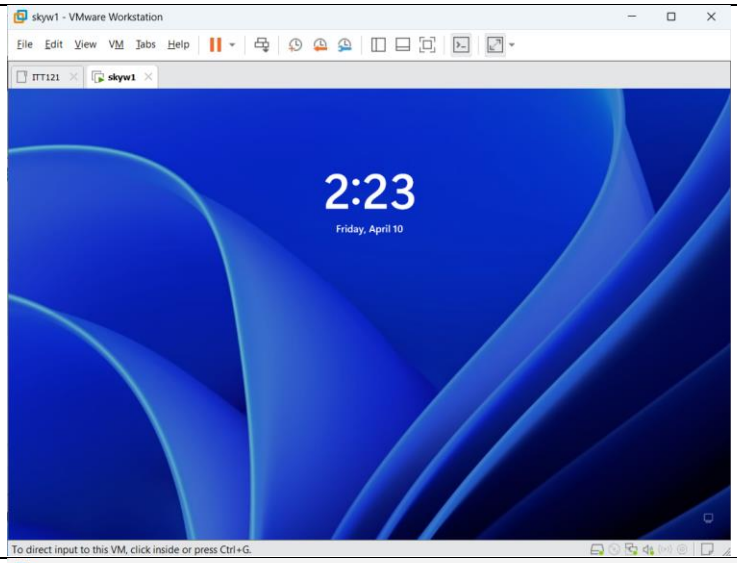
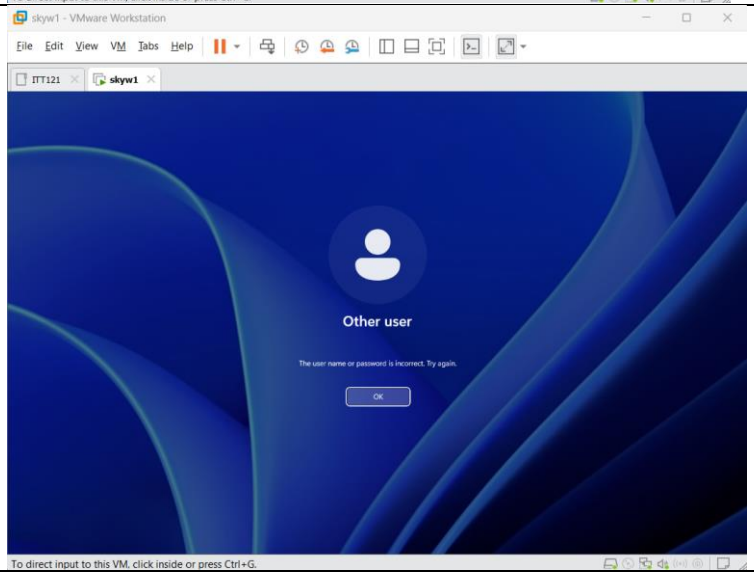
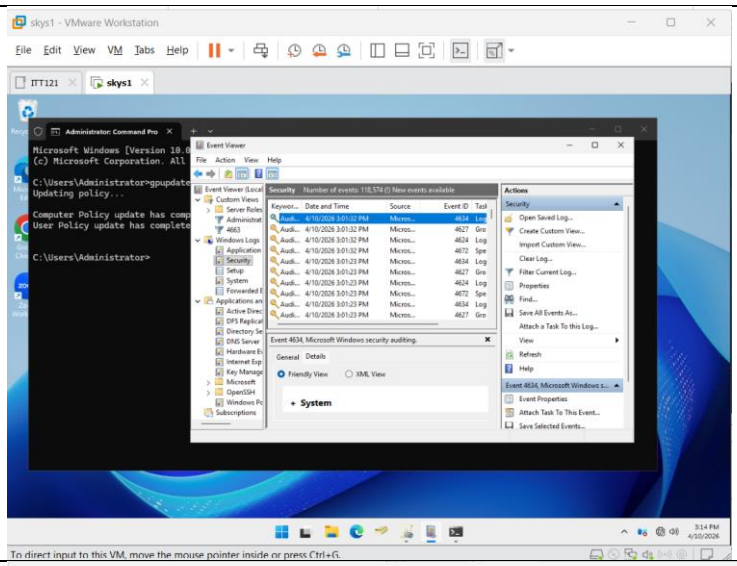
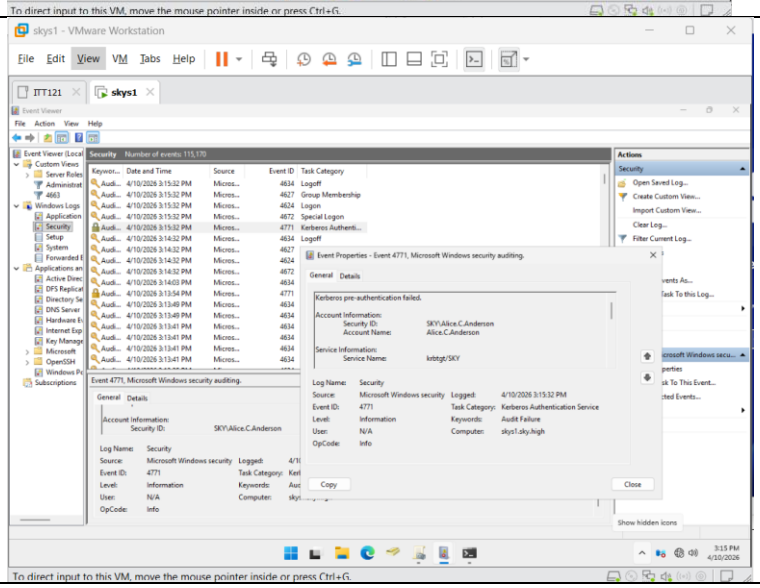
To start auditing common threats, we need to make a new group policy in the server. In your server, open group policy management and right click the domain name and create a new GPO and link it here.	
Give a name and then right click the new GPO and click edit to open the editing screen. Follow this path to get to the auditing screen. Computer Configuration -> Polocies -> Windows Settings -> Security Settings -> Local Policies	
Now we need to go down the list and configure each policy. Starting from the top with account logon events, double click to open properties and check the “define these policy settings” checkbox and also check the success and failure boxes. Hit apply and ok.	

Now move to the next policy, account management. Enable the policy and check the success and failure boxes. Hit apply and then ok.	
For the next policy, service access, define the policy and only check success. Hit apply and ok.	
For the next policy, logon events, define the policy and then check success and failure. Hit apply and ok.	

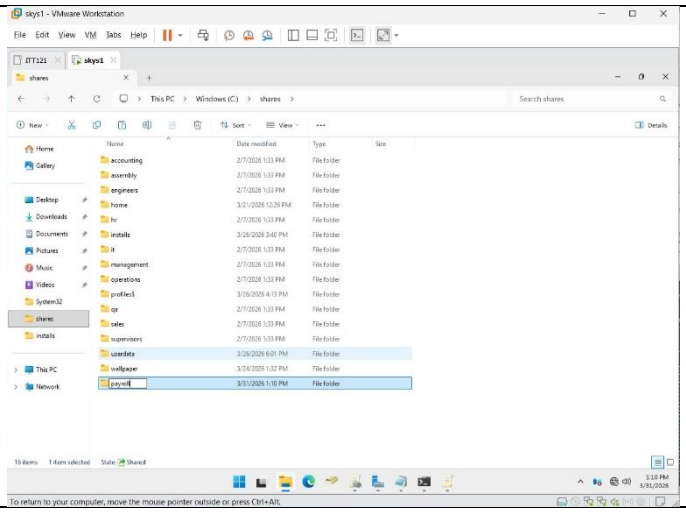
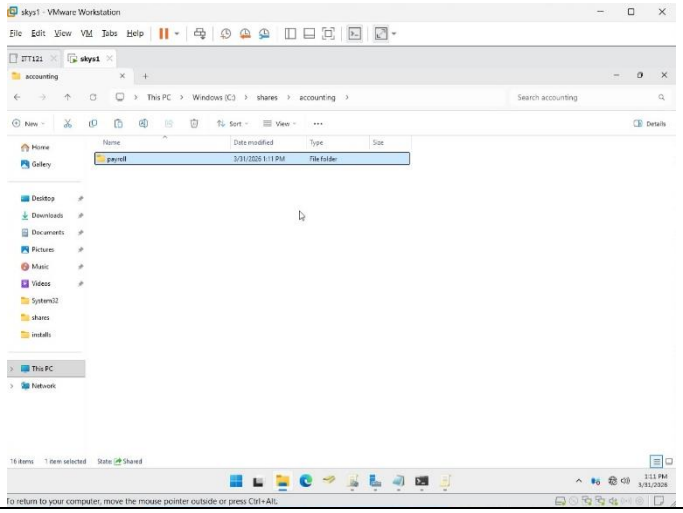
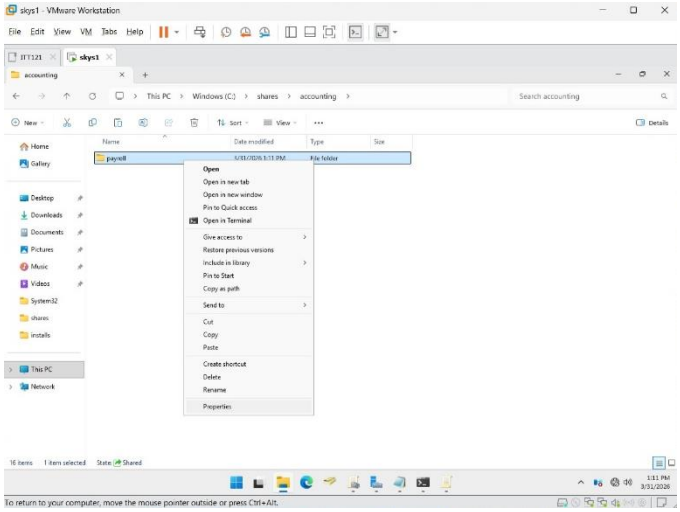
For object access, we don't have to define anything and we can move to the next policy.	
For policy change, define the policy and check success. Click apply and ok.	
For privilege use, we do not need to define anything and we can skip this policy.	

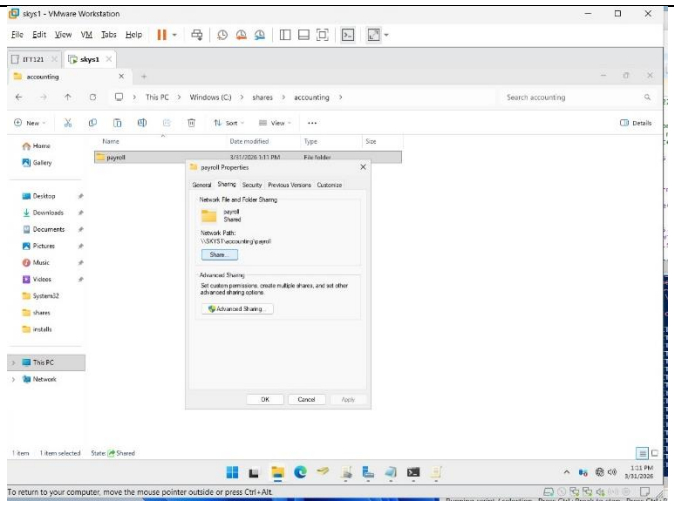
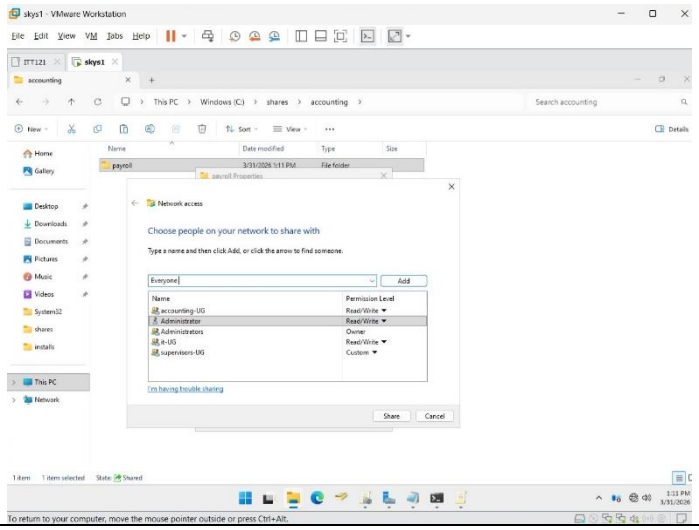
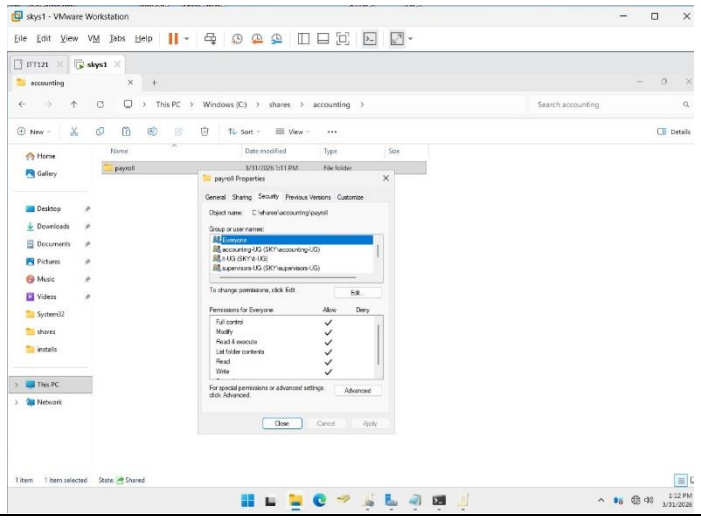
For process tracking define the policy and check success. Hit apply and then ok.	
For the final policy, system events, define the policy and check the success box, the hit apply and ok.	
Now we need to apply the group policy updates. Click the windows icon and type “cmd”. Then in the terminal type out “gpupdate /force” and then hit enter to apply the new group policy settings.	

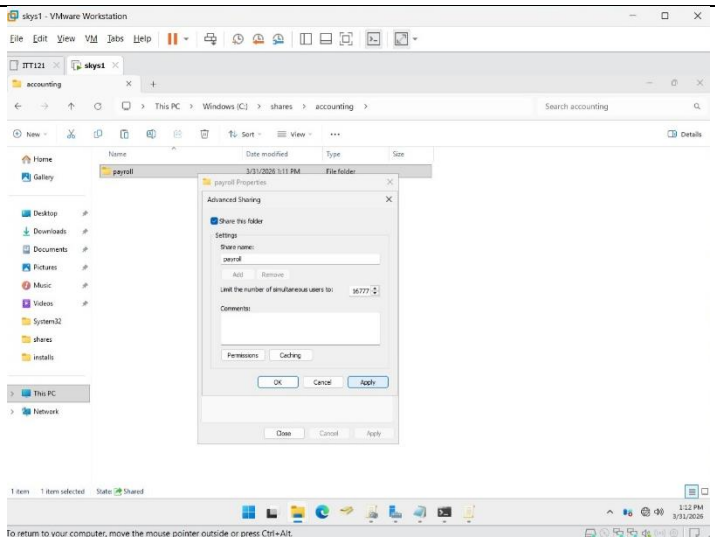
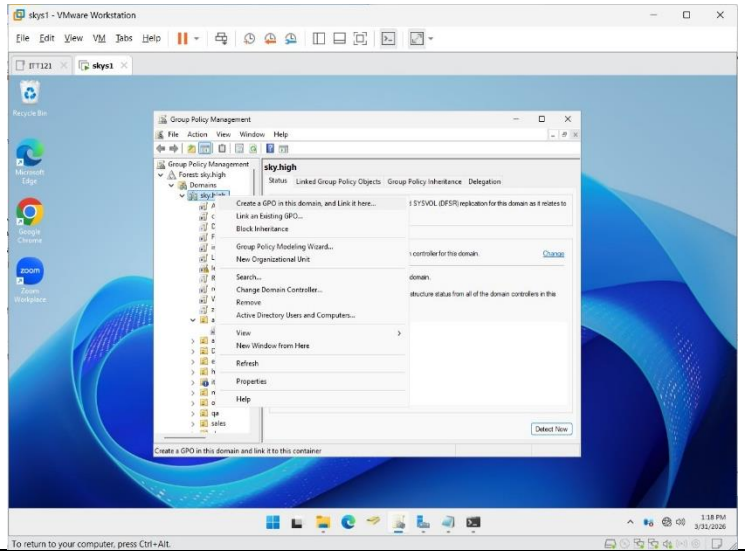
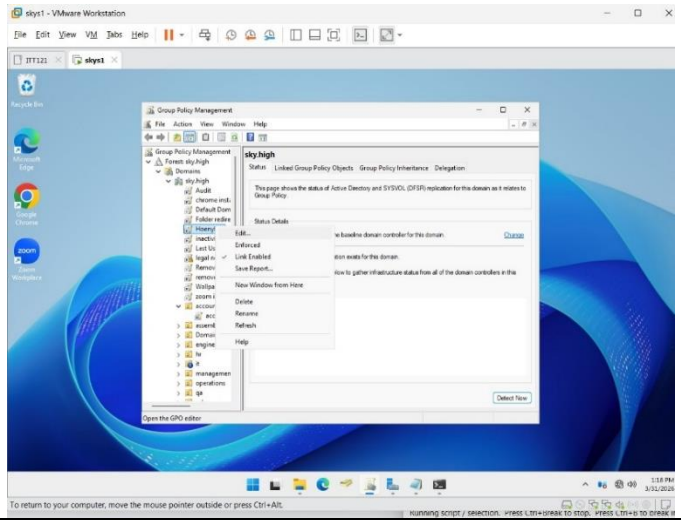
Now on the workstation signed in as it (because GP inheritance is disabled for it) click the windows icon and type in cmd and then hit enter to open command prompt. Type out “gpupdate /force” and then hit enter.	
Now sign out and pick a new user to sign in. We are going to simulate a brute force password attack and test our new auditing settings. Pick a user to sign as.	
Now try to sign in with the wrong password a couple times, if the auditing settings are configured we should be able to see logs for failed logon events.	

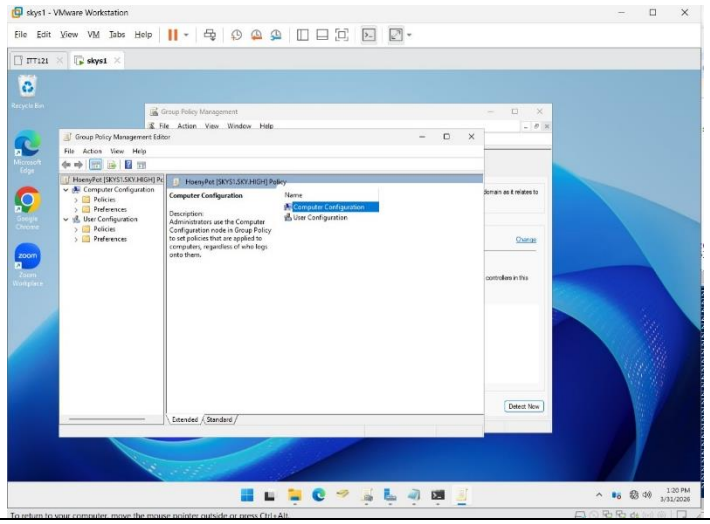
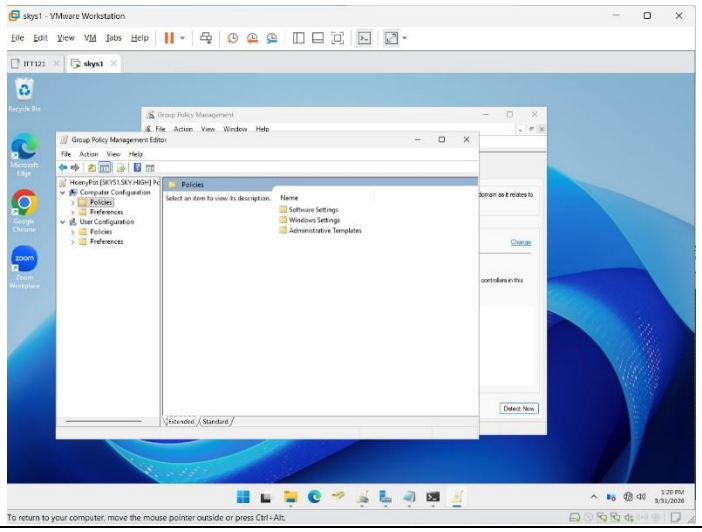
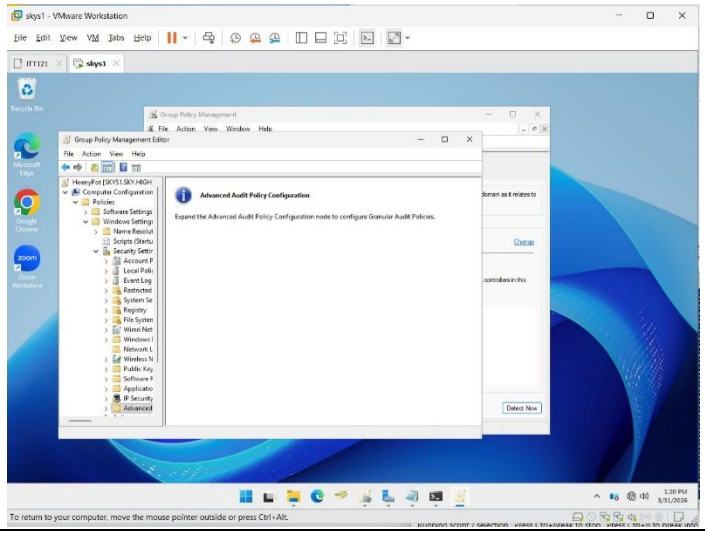
Back in the server open event viewer, and under windows logs open the security page.	
Look for id 4771 and we can see that this event is a failed authentication event for the user we tried to sign in as. If we saw large number of these event logs it could mean that one of our users accounts is being hit with a brute force attack.	

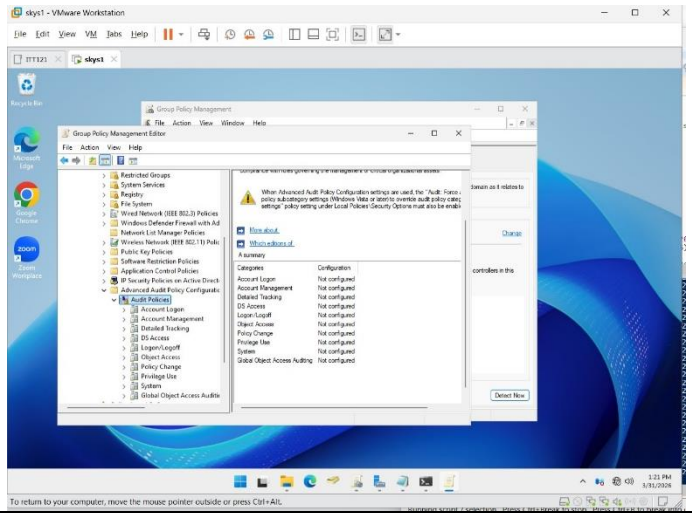
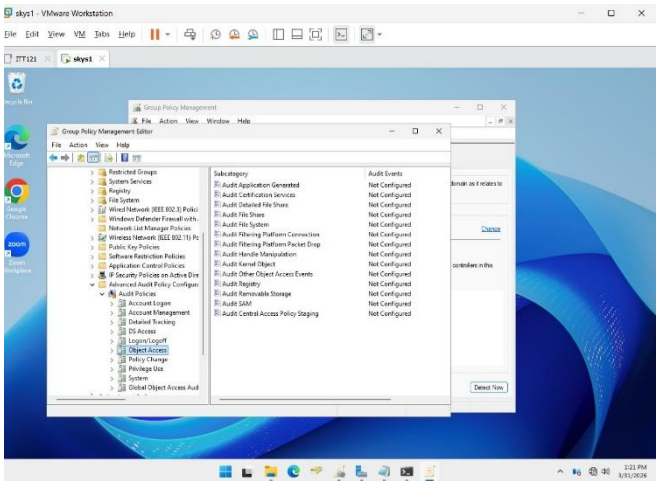
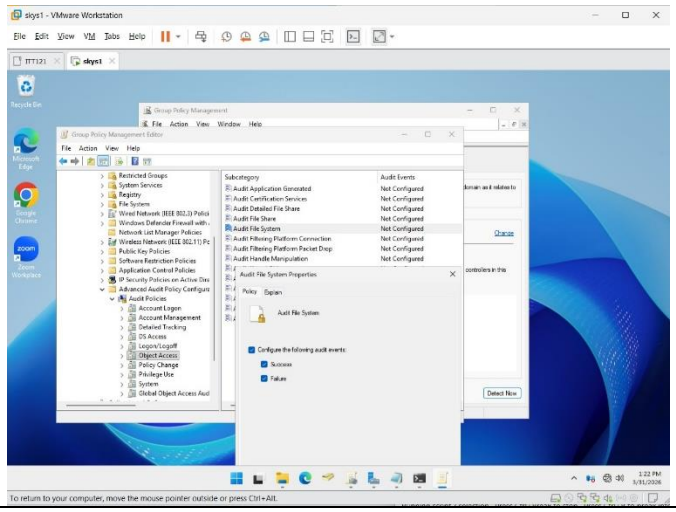
To start creating out honeypot we need to make an attractive folder in our shares.	

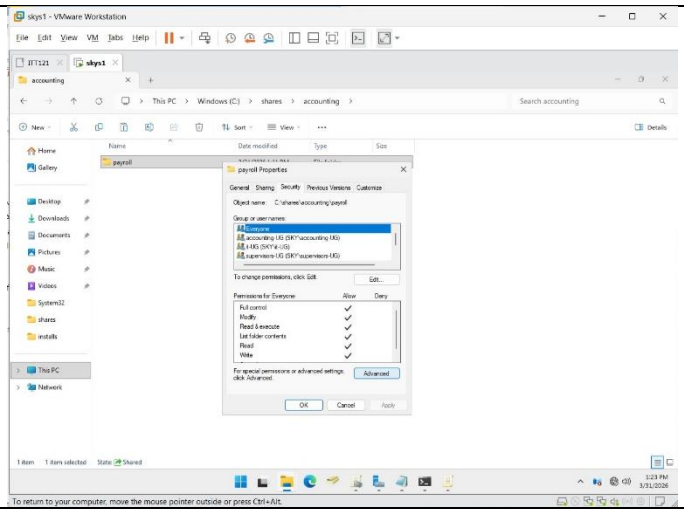
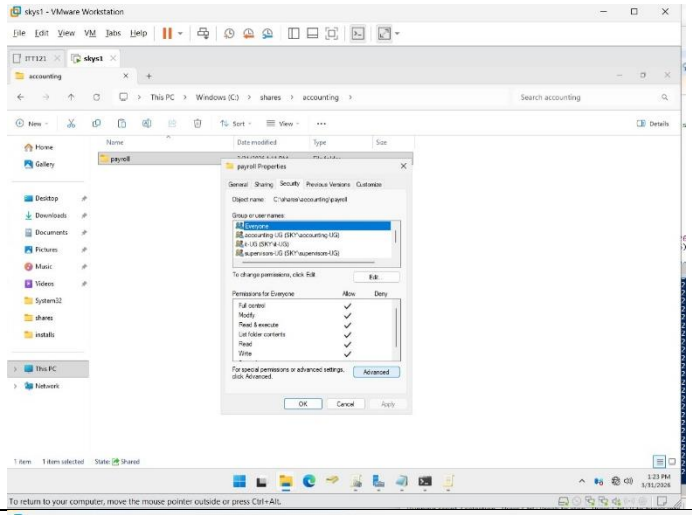
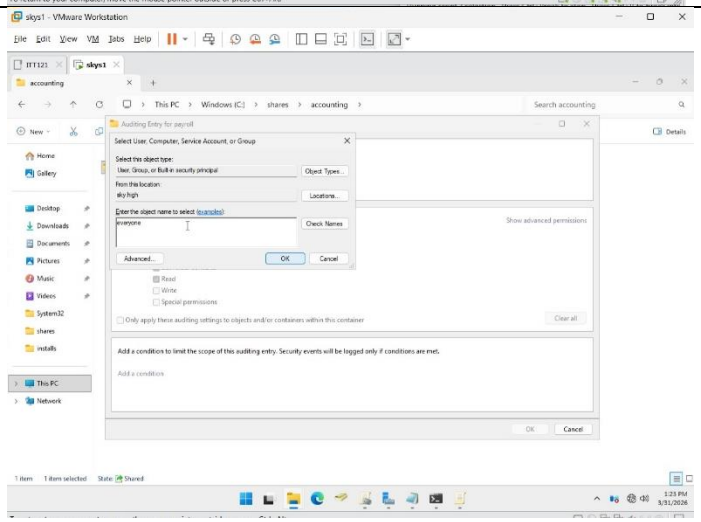
	
Navigate shares folder on the server and make a new folder called “payroll” inside accounting’s department folder.	
Now we need to configure the sharing and security properties for the folder so right click the new folder and click on properties	

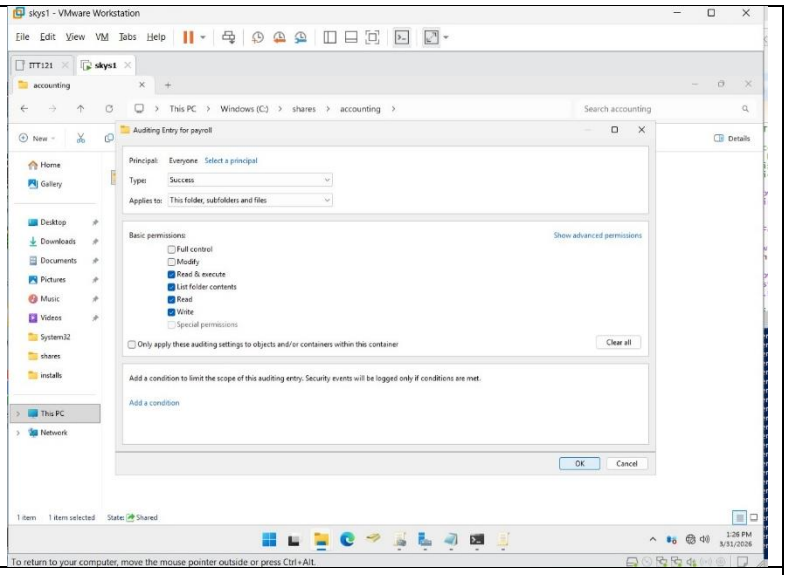
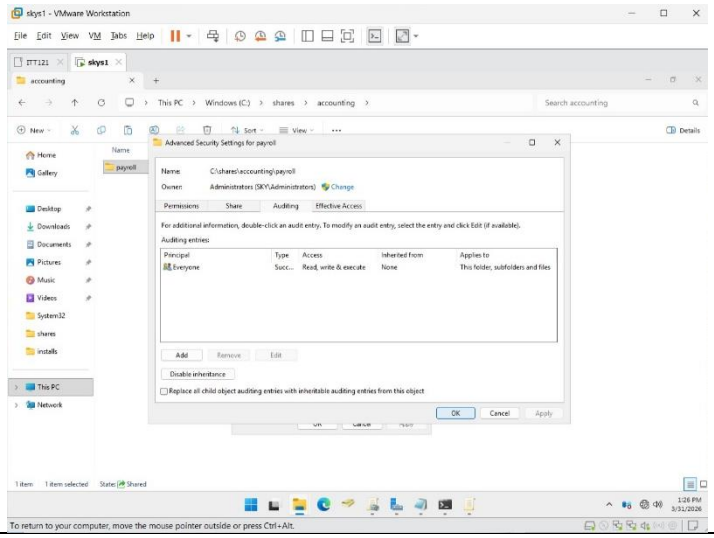
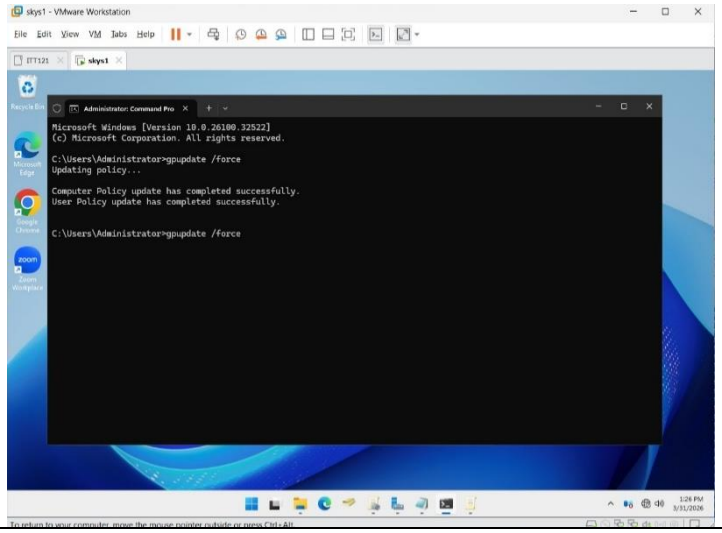
Go to the sharing tab and open up advanced sharing	
Type out Everyone and hit enter to add the user group to the folder permissions.	
Make sure that everyone has full control	

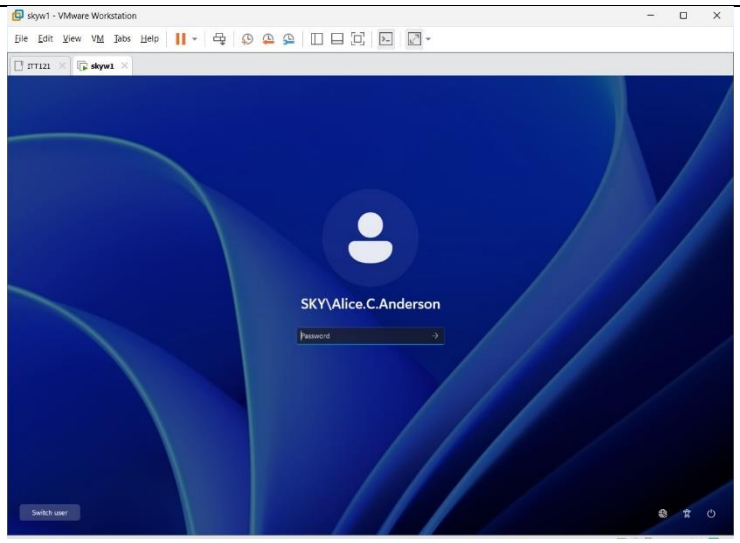
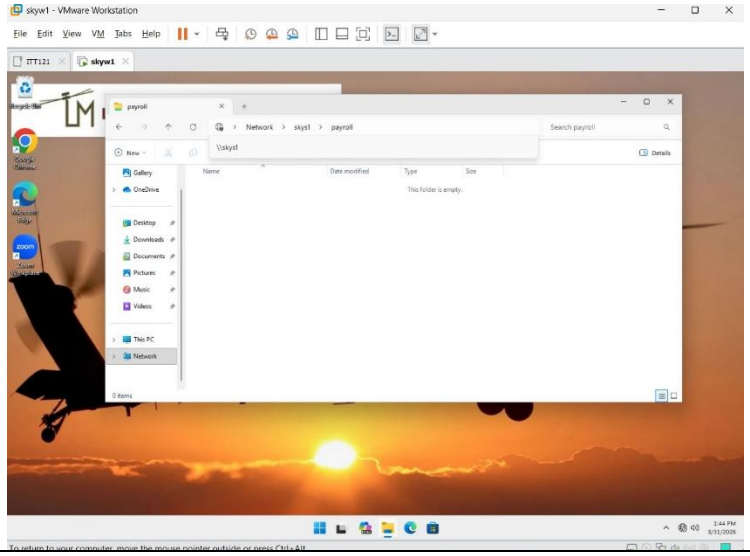
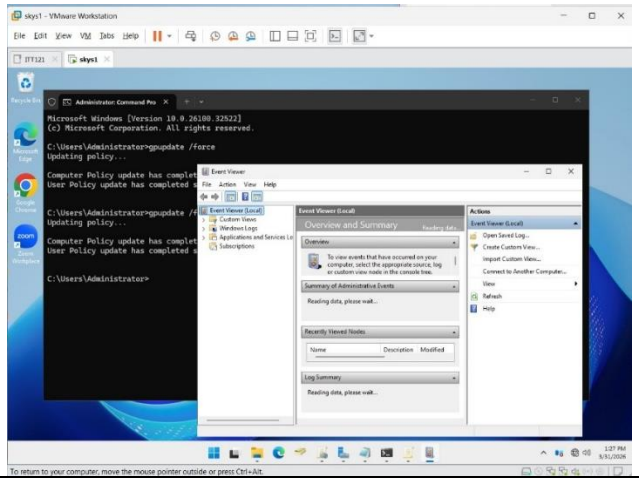
In the folder's sharing tab, hit the share button and make sure that "share this folder" is checked. Hit apply and then okay.	
Now on the server, go to Group Policy Management and right click the domain name to create a new GPO and link it.	
Name the new GPO Honeypot and then right click the new GPO and find the edit option	

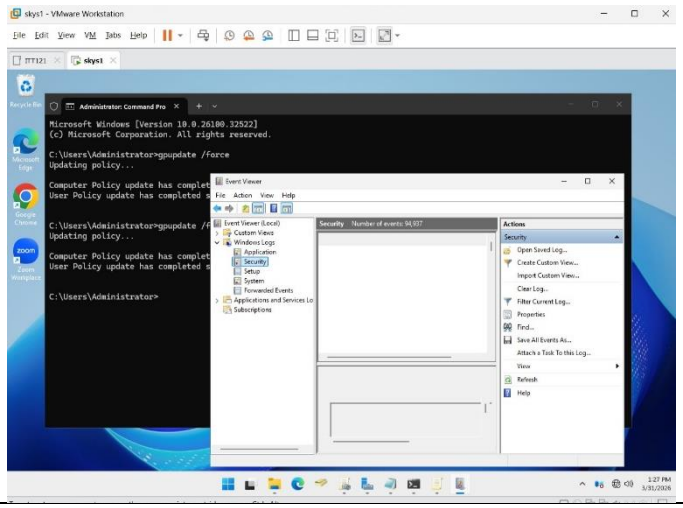
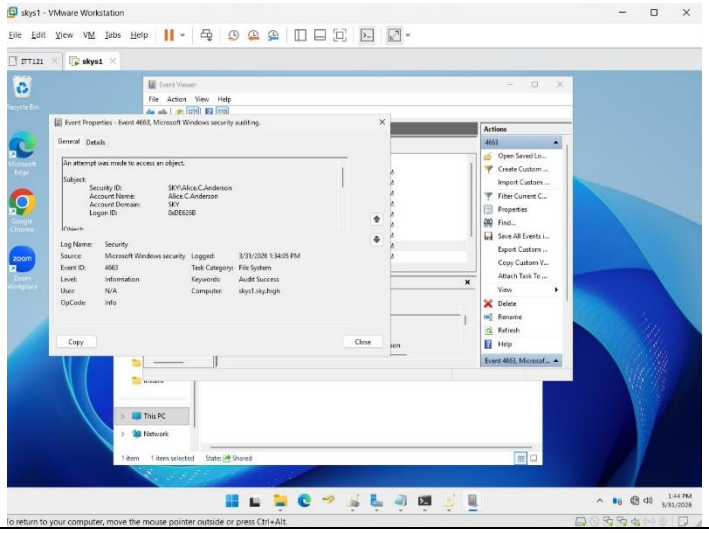
Now go to Windows Configuration once inside the edit window	
Then go into Windows Settings	
Now to get to Advanced Audit Policy Configuration and the branches inside, we need to expand the folder on the left hand side	

Once expanded, expand the Audit Policies folder	
Now find the option “Object Access” and click on the category, then in sub categories, click on Audit File Systems	
Check the box to configure the auditing events and select the boxes for success and failure	
Now in the server go back to the payroll folder and open the	

properties again by right clicking on the folder	
Find the security tab and then select the advanced options. Once in advanced options find the auditing tab	
Open the Auditing Entry and you should see that everything is blocked out. At the top click the blue link that says “select a principle”. Now in the text box type out everyone and hit okay.	
Make sure that everyone has read and write privileges.	

	
Once you are done hit okay and now the principle should be added in the Auditing tab for advanced security. Hit apply and then ok.	
Now to apply our changes open command prompt by hitting the windows icon and searching cmd and then hitting the enter key. Now in the terminal type out “gpupdate /force” and hit enter.	

Now find an accounting user to log into the workstation with.	
Inside the workstation as an Accounting user, open file explorer and in the search bar, type out \\servername\accounting\payroll	
Now back in the server, click on the windows icon and search for event viewer.	

Once open navigate into Windows Logs, then security.	
Give it a second to load the logs and then find or filter for the event ID 4663. Once you find an event with the matching ID, click on it and you should see details about the event. You should see details about who tried to access the honeypot as well and it should be the name of the user we are signed into on the workstation.	

Auditing is a critical component of cybersecurity defense at Lopes Manufacturing because it provides visibility into user activity, system access, and potential threats to sensitive data, such as proprietary helicopter part designs and military-related information. By implementing auditing policies and monitoring event logs such as file access events 4656 and 4663, the organization can detect unauthorized access attempts, insider threats, or suspicious behavior on its network. For a honeypot, auditing allows security teams to track exactly who accessed the resource, what actions were taken, and when they occurred, turning a potential vulnerability into a detection mechanism. However, a proper honeypot deployment in a corporation like Lopes Manufacturing requires more than just auditing. Additional tasks could include network segmentation to isolate the honeypot from critical systems, alerting mechanisms for real-time threat detection, regular log analysis or integration with a SIEM solution, strict access controls, and incident response procedures. Together, these measures ensure that the

honeypot not only detects malicious activity but also supports a broader, proactive cybersecurity strategy that protects mission-critical operations and sensitive military data.

References

What is a honeypot. (2025). GeeksforGeeks. https://www.geeksforgeeks.org/blogs/what-is-honeypot/

Viadeeswaran, N. (2025). *Honeypots in cybersecurity explained.* CrowdStrike. http://crowdstrike.com/en-us/